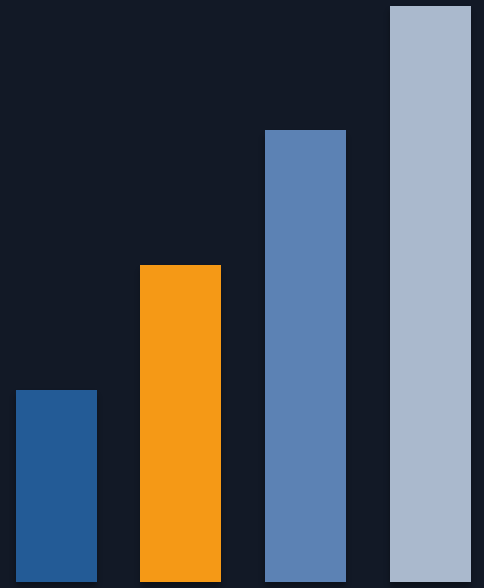


E-COMMERCE SQL ANALYTICS



Q00

Project dashboard

A complete view of the 14 questions included in the source project

Q1–Q4	**Sales & Customer Value**	Seller-state orders • cumulative revenue • payment behavior • highest spender
Q5–Q8	**Products & Marketplace**	Category ratings • customer orders • inactive sellers • top revenue categories
Q9–Q11	**Operations**	Median delivery • never-ordered products • high-performing sellers
Q12–Q14	**Experience & Trends**	Highest review-score state • no-review customers • peak order month

Note: Results below reproduce the result values visible in the uploaded project. Where the source shows only sample rows, the slide labels them as displayed rows.

Q01

Orders Fulfilled by Seller State

Question • Result • Business takeaway • SQL solution

BUSINESS QUESTION

Find the total number of orders fulfilled by each seller state.

SQL ANSWER

```
SELECT seller_state, count(DISTINCT order_id) AS total_order
FROM order_items
JOIN sellers USING (seller_id)
GROUP BY seller_state;
```

seller_state	total_order
AC	1
AM	3
BA	569
CE	91
DF	824
ES	318

Source: uploaded Amazon E-Commerce SQL Analytics project

Q02

Cumulative Revenue over Purchase Time

Question • Result • Business takeaway • SQL solution

BUSINESS QUESTION

For each product category, calculate the cumulative revenue generated as orders come in over time.

SQL ANSWER

```
SELECT order_purchase_timestamp, product_category_name,
ROUND(SUM(price+freight_value), 2) AS revenue,
ROUND(SUM(SUM(price+freight_value))
OVER(PARTITION BY product_category_name ORDER BY
order_purchase_timestamp DESC), 2) AS cumulative_total
FROM order_items JOIN products USING (product_id)
JOIN orders USING (order_id)
GROUP BY product_category_name, order_purchase_timestamp;
```

RESULT

order_purchase_timestamp	product_category_name	revenue	cumulative_total
2018-04-28 23:14:00	agro_industria_e_come...	104.44	21659.73
2018-04-27 08:44:30	agro_industria_e_come...	433.46	22093.19
2018-04-25 17:53:32	agro_industria_e_come...	244.48	22337.67
2018-04-24 22:53:20	agro_industria_e_come...	100.12	22437.79
2018-04-19 18:02:15	agro_industria_e_come...	247.48	22685.27
2018-04-18 16:33:30	agro_industria_e_come...	1518.55	24203.82

Source: uploaded Amazon E-Commerce SQL Analytics project

Q03

Most Used Payment Method & Average Order Value

Question • Result • Business takeaway • SQL solution

BUSINESS QUESTION

Which payment method do customers use the most, and what is the average order value for each payment type?

SQL ANSWER

```
SELECT payment_type,
COUNT(payment_type) AS total_orders,
ROUND(AVG(payment_value), 2) AS avg_order_value
FROM order_payments
GROUP BY payment_type
ORDER BY total_orders DESC;
```

payment_type	total_orders	avg_order_value
credit_card	76795	163.32
boleto	19784	145.03
voucher	5775	65.7
debit_card	1529	142.57
not_defined	3	0

Source: uploaded Amazon E-Commerce SQL Analytics project

Q04

Highest Spending Customer

Question • Result • Business takeaway • SQL solution

BUSINESS QUESTION

Find the customer who has spent the most money across all their orders.

SQL ANSWER

```
SELECT customer_id,
ROUND(SUM(price+freight_value), 2) AS customer_spent,
ROW_NUMBER() OVER(ORDER BY ROUND(SUM(price+freight_value), 2) DESC)
AS rank_by_spent
FROM orders
JOIN order_items USING (order_id)
GROUP BY customer_id
ORDER BY customer_spent DESC;
```

customer_id	customer_spent	rank_by_spent
1617b1357756262bfa56ab541c47bc16	13664.08	1
ec5b2ba62e574342386871631fafd3fc	7274.88	2
c6e2731c5b391845f6800c97401a43a9	6929.31	3
f48d464a0baaea338cb25f816991ab1f	6922.21	4
3fd6777bbce08a352fddd04e4a7cc8f6	6726.66	5
05455dfa7cd02f13d132aa7a6a9729c6	6081.54	6

Source: uploaded Amazon E-Commerce SQL Analytics project

Q05

Average Review Score per Product Category

Question • Result • Business takeaway • SQL solution

BUSINESS QUESTION

Find the average review score for each product category.

SQL ANSWER

```
SELECT product_category_name,
ROUND(AVG(review_score), 2) AS avg_rating
FROM products
JOIN order_items USING (product_id)
JOIN order_reviews USING (order_id)
GROUP BY product_category_name;
```

product_category_name	avg_rating
relogios_presentes	4.02
livros_tecnicos	4.37
informatica_acessorios	3.93
eletronicos	4.04
esporte_lazer	4.11
bebidas	4.05

Source: uploaded Amazon E-Commerce SQL Analytics project

Q06

Total Orders by Customer and State

Question • Result • Business takeaway • SQL solution

BUSINESS QUESTION

Find the total number of orders placed by each customer, broken down by the state they live in.

SQL ANSWER

```
SELECT customer_id, customer_state,
COUNT(DISTINCT order_id) AS total_orders
FROM customers
JOIN orders USING (customer_id)
GROUP BY customer_id, customer_state;
```

customer_id	customer_state	total_orders
00012a2ce6f8dcda20d059ce98491703	SP	1
000161a058600d5901f007fab4c27140	MG	1
0001fd6190edaaf884bcaf3d49edf079	ES	1
0002414f95344307404f0ace7a26f1d5	MG	1
000379cdec625522490c315e70c7a9fb	SP	1
0004164d20a9e969af783496f3408652	SP	1
000419c5494106c306a97b5635748086	RJ	1

Source: uploaded Amazon E-Commerce SQL Analytics project

Q07

Unregistered or Inactive Sellers

Question • Result • Business takeaway • SQL solution

BUSINESS QUESTION

Identify sellers who registered on the platform but have never fulfilled a single order.

SQL ANSWER

```
SELECT s.seller_id,seller_state
FROM sellers s
WHERE NOT EXISTS (
SELECT 1
FROM order_items oi
WHERE s.seller_id = oi.seller_id
);
```

seller_id	seller_state
3442f8959a84dea7ee197c632cb2df15	SP
d1b65fc7debc3361ea86b5f14c68d2e2	SP
ce3ad9de960102d0677a81f5d0bb7b2d	RJ
c0f3eea2e14555b6faeea3dd58c1b1c3	SP

Source: uploaded Amazon E-Commerce SQL Analytics project

Q08

Top 5 Product Categories by Revenue

Question • Result • Business takeaway • SQL solution

BUSINESS QUESTION

Find the top 5 product categories by total revenue.

SQL ANSWER

```
SELECT * FROM
(
SELECT product_category_name,
ROUND(SUM(price + freight_value), 2) AS revenue,
DENSE_RANK() OVER
(ORDER BY ROUND(SUM(price + freight_value), 2) DESC) AS product_rank
FROM products
JOIN order_items USING (product_id)
GROUP BY product_category_name
) t
WHERE product_rank <= 5;
```

product_category_name	revenue	product_rank
beleza_saude	1441248.07	1
relogios_presentes	1305541.61	2
cama_mesa_banho	1241681.72	3
esporte_lazer	1156656.48	4
informatica_acessorios	1059272.4	5

Source: uploaded Amazon E-Commerce SQL Analytics project

Q09

Median Delivery Time

Question • Result • Business takeaway • SQL solution

BUSINESS QUESTION

Find the median delivery time in days between order placement and actual delivery.

SQL ANSWER

```
WITH OrderedTime AS (
SELECT DATEDIFF(order_delivered_customer_date,
order_purchase_timestamp) AS delivery_time_in_days,
ROW_NUMBER() OVER(ORDER BY DATEDIFF(
order_delivered_customer_date, order_purchase_timestamp)) AS row_num,
COUNT(*) OVER() AS total_count
FROM orders
WHERE order_delivered_customer_date IS NOT NULL
AND order_purchase_timestamp IS NOT NULL
)
SELECT AVG(delivery_time_in_days) AS median_delivery_time
FROM OrderedTime
WHERE row_num IN (FLOOR((total_count + 1) / 2),
CEIL((total_count + 1) / 2));
```

median_delivery_time
10.0000

Source: uploaded Amazon E-Commerce SQL Analytics project

Q10

Products Never Ordered

Question • Result • Business takeaway • SQL solution

BUSINESS QUESTION

Find all products that have never been ordered.

SQL ANSWER

```
SELECT product_id,product_category_name
FROM products p
WHERE NOT EXISTS (
SELECT 1 FROM order_items oi
WHERE p.product_id = oi.product_id
);
```

product_id	product_category_name
1e9e8ef04dbcff4541ed26657ea517e5	perfumaria
3aa071139cb16b67ca9e5dea641aaa2f	artes
96bd76ec8810374ed1b65e291975717f	esporte_lazer
cef67bcfe19066a932b7673e239eb23d	bebes
9dc1a7de274444849c219cff195d0b71	utilidades_domesticas
41d3672d4792049fa1779bb35283ed13	instrumentos_musicais

Source: uploaded Amazon E-Commerce SQL Analytics project

Q11

High-Performing Sellers

Question • Result • Business takeaway • SQL solution

BUSINESS QUESTION

Find sellers who have fulfilled more orders than the average seller on the platform.

SQL ANSWER

```
WITH avgCte AS (
SELECT seller_id, COUNT(order_id) AS total_orders
FROM orders
JOIN order_items USING (order_id)
GROUP BY seller_id
)
SELECT seller_id, total_orders
FROM avgCte
WHERE total_orders >
(SELECT AVG(total_orders) FROM avgCte);
```

seller_id	total_orders
41b39e28db005d9731d9d485a83b4c38	231
4a3ca9315b744ce9f8e9374361493884	1987
0241d4d5d36f10f80c644447315af0bd	246
516e7738bd8f735ac19a010ee5450d8d	84
7c67e1448b00f6e969d365cea6b010ab	1364
fe2032dab1a61af8794248c8196565c9	306
cab85505710c7cb9b720bceb52b01cee	209

Source: uploaded Amazon E-Commerce SQL Analytics project

Q12

Brazilian States with Highest Review Scores

Question • Result • Business takeaway • SQL solution

BUSINESS QUESTION

Find which Brazilian states have the highest average customer review score for orders delivered there.

SQL ANSWER

```
SELECT customer_state, ROUND(AVG(review_score), 2) AS avg_score
FROM customers
JOIN orders USING (customer_id)
JOIN order_items USING (order_id)
JOIN order_reviews USING (order_id)
GROUP BY customer_state
ORDER BY avg_score DESC LIMIT 1;
```

customer_state	avg_score
AP	4.22

Source: uploaded Amazon E-Commerce SQL Analytics project

Q13

Customers with Orders but No Reviews

Question • Result • Business takeaway • SQL solution

BUSINESS QUESTION

Identify customers who have placed orders but never left a review.

SQL ANSWER

```
SELECT customer_id, review_comment_message
FROM orders
JOIN order_reviews USING (order_id)
WHERE review_comment_message IS NULL;
```

customer_id	review_comment_message
41ce2a54c0b03bf3443c3d931a367089	NULL
be8c14c16a4d47194ccdfe10f1fc5b1a	NULL
0bf19317b1830a69e55b40710576aa7a	NULL
9a4900710ff76d73cb3a0d152e73adf0	NULL
f3f0e613e0bdb9c7cee75504f0f90679	NULL
c572fd6cdd98e87ea2bc433b6da21078	NULL
2528510465c7572dedee4431a7d52e08	NULL

Source: uploaded Amazon E-Commerce SQL Analytics project

Q14

Month with the Highest Orders

Question • Result • Business takeaway • SQL solution

BUSINESS QUESTION

Find the month with the highest number of orders placed across the entire platform.

SQL ANSWER

```
SELECT MONTH(order_purchase_timestamp) AS MONTH,
YEAR(order_purchase_timestamp) AS YEAR,
COUNT(DISTINCT order_id) AS TOTAL_ORDERS,
RANK() OVER(ORDER BY count(DISTINCT order_id) DESC) AS ORDER_RANK
FROM orders
GROUP BY MONTH(order_purchase_timestamp),
YEAR(order_purchase_timestamp)
LIMIT 1;
```

MONTH	YEAR	TOTAL_ORDERS	ORDER_RANK
11	2017	7544	1

Source: uploaded Amazon E-Commerce SQL Analytics project

Thank you

Presented by: Anurag Sharma